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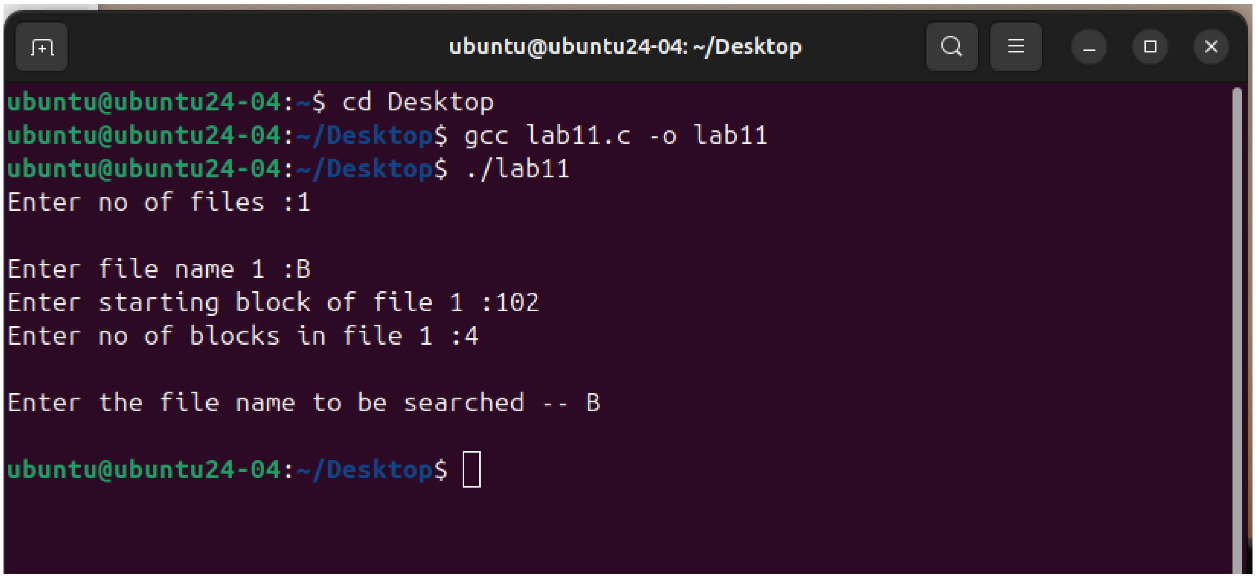
6B2

Sample

```
#include <stdio.h>
#include<string.h>
struct fileTable
{
   char name[20];
   int sb, nob;
} ft[30];

void main()
{
   int i, j, n;
   char s[20];

   printf("Enter no of files :");
   scanf("%d", &n);

   for (i = 0; i < n; i++)
   {
      printf("\nEnter file name %d :", i + 1);
      scanf("%s", ft[i].name);

      printf("Enter starting block of file %d :", i + 1);
      scanf("%d", &ft[i].sb);

      printf("Enter no of blocks in file %d :", i + 1);
      scanf("%d", &ft[i].nob);
   }

   printf("\nEnter the file name to be searched -- ");
   scanf("%s", s);

   for (i = 0; i < n; i++)
      if (strcmp(s, ft[i].name) == 0)
         break;

   if (i == n)
      printf("\nFILE NAME OCCUPIED\n");
```

```
    printf("\n%s\t\t%d\t\t%d\t", ft[i].name, ft[i].sb, ft[i].nob);

    for (j = 0; j < ft[i].nob; j++)
        printf("%d, ", ft[i].sb + j);
}
```

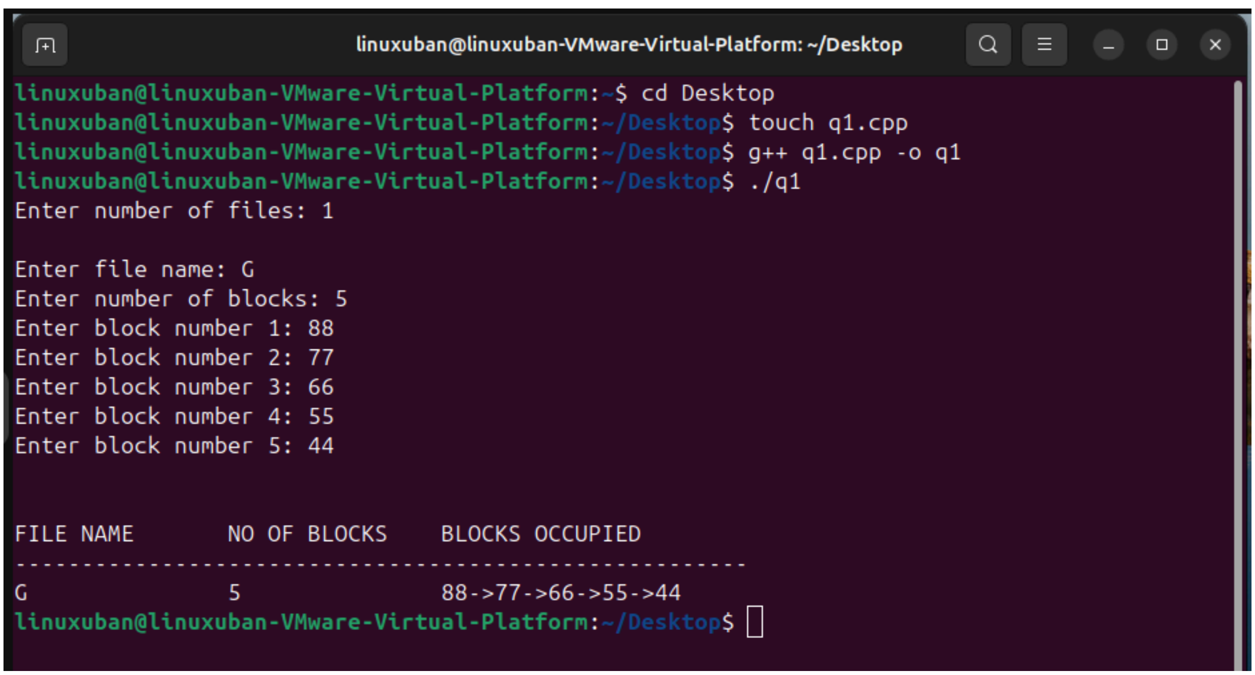


Q1

```
#include <iostream>
#include <cstring>
using namespace std;

// Structure for each disk block
struct block
{
    int bno;
    block *next;
};

// Structure for file table
struct fileTable
{
    char name[20];
    int nob;        // number of blocks
```

```
    block *sb;     // starting block
} ft[30];

int main()
{
    int n; // number of files

    cout << "Enter number of files: ";
    cin >> n;

    for (int i = 0; i < n; i++)
    {
        cout << "\nEnter file name: ";
        cin >> ft[i].name;

        cout << "Enter number of blocks: ";
        cin >> ft[i].nob;

        block *temp, *newBlock;

        for (int j = 0; j < ft[i].nob; j++)
        {
            newBlock = new block;

            cout << "Enter block number " << j + 1 << ": ";
            cin >> newBlock->bno;
            newBlock->next = NULL;

            if (j == 0)
            {
                ft[i].sb = newBlock; // first block
                temp = newBlock;
            }
            else
            {
                temp->next = newBlock; // link previous block
                temp = newBlock;
            }
        }
    }

    // Display file allocation table
    cout << "\n\nFILE NAME\tNO OF BLOCKS\tBLOCKS OCCUPIED\n";
    cout << "--------------------------------------------------------\n";
```

```
  for (int i = 0; i < n; i++)
  {
    cout << ft[i].name << "\t\t" << ft[i].nob << "\t\t";

    block *temp = ft[i].sb;

    while (temp != NULL)
    {
      cout << temp->bno;
      if (temp->next != NULL)
        cout << "->";
      temp = temp->next;
    }

    cout << endl;
  }

  return 0;
}
```



Q2

```cpp
#include <iostream>
#include <cstring>
using namespace std;

// Structure for file table
struct fileTable
{
    char name[20];
    int nob;            // Number of blocks
    int blocks[30];     // Indexed blocks
} ft[30];

int main()
{
    int n; // Number of files

    cout << "Enter number of files: ";
    cin >> n;

    // Input file details
    for (int i = 0; i < n; i++)
    {
        cout << "\nEnter file name: ";
        cin >> ft[i].name;

        cout << "Enter number of blocks: ";
        cin >> ft[i].nob;

        cout << "Enter block numbers:\n";
        for (int j = 0; j < ft[i].nob; j++)
        {
            cout << "Block " << j + 1 << ": ";
            cin >> ft[i].blocks[j];
        }
    }

    // Display Indexed File Allocation Table
    cout << "\n\nFILE NAME\tNO OF BLOCKS\tBLOCKS OCCUPIED\n";
    cout << "--------------------------------------------------\n";

    for (int i = 0; i < n; i++)
    {
        cout << ft[i].name << "\t\t" << ft[i].nob << "\t\t";
```

```
        for (int j = 0; j < ft[i].nob; j++)
        {
            cout << ft[i].blocks[j];
            if (j != ft[i].nob - 1)
                cout << ", ";
        }

        cout << endl;
    }

    return 0;
}
```